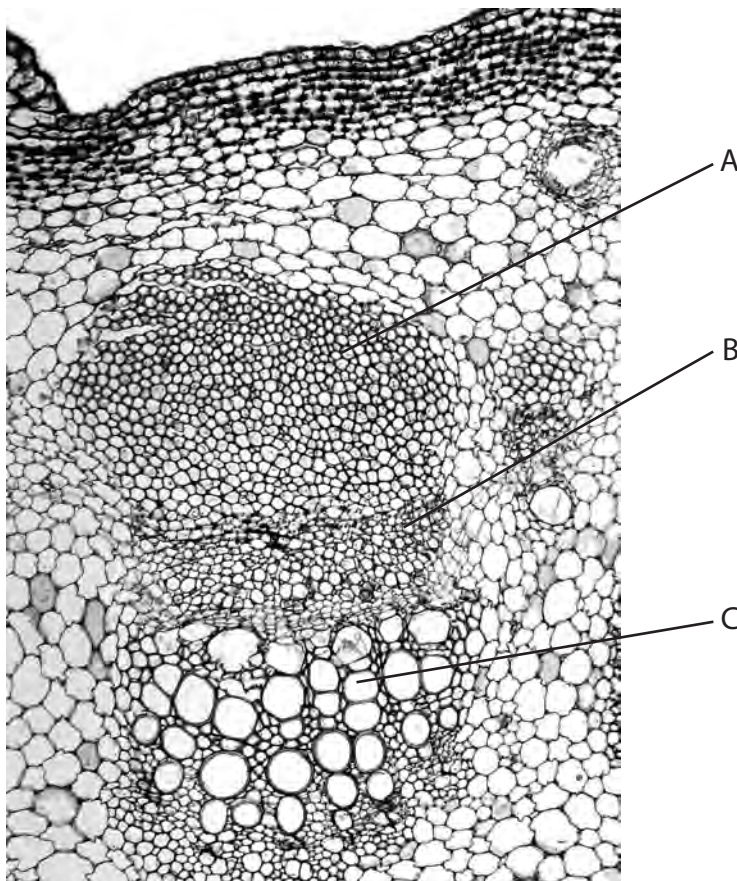


Answer ALL questions.

Write your answers in the spaces provided.

- 1** A student used a microscope to study the tissues in a plant stem and the cell of an animal.

The photograph shows a vascular bundle in a transverse section of the plant stem, as seen using a light microscope.



(Source: © DR KEITH WHEELER/SCIENCE PHOTO LIBRARY)

- (a) (i) Describe how you would use the information in the photograph to identify structures A and C.

(2)

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- (ii) A longitudinal section was taken through one of the labelled tissues A, B or C. The photograph shows a view of some of the cells in this tissue, as seen using a light microscope.



(Source: © DR KEITH WHEELER/SCIENCE PHOTO LIBRARY)

Deduce which of the tissues A, B or C these cells come from.

(3)

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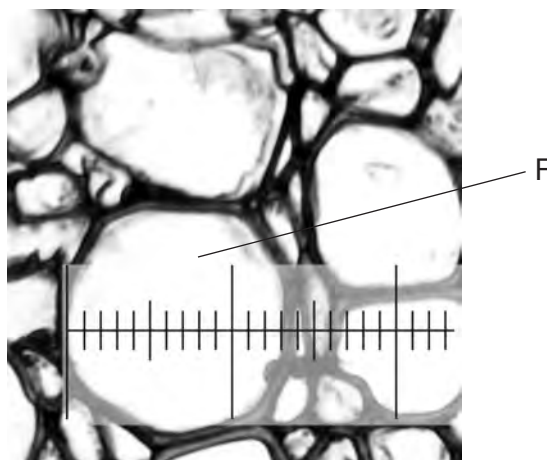
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- (iii) This photograph shows a high-power view of some of the cells in tissue C in the vascular bundle.

Each of the smallest units on the eyepiece graticule shown in the photograph is $3 \times 10^{-6} \text{ m}$.



(Source: © DR KEITH WHEELER/SCIENCE PHOTO LIBRARY)

Assume that the cell labelled F in the photograph is circular.

Calculate the cross-sectional area of cell F.

(2)

Answer

- (iv) A cell in tissue B of the vascular bundle has a cross-sectional area of $6.13 \times 10^{-11} \text{ m}^2$.

Calculate how many times larger the cross-sectional area of cell F is than this cell in tissue B.

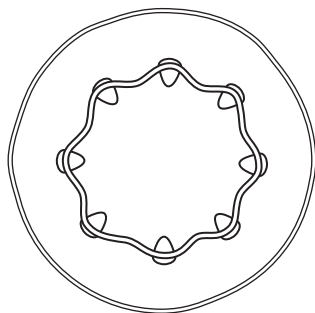
Give your answer to **three** significant figures.

(1)

Answer



- (v) In another part of the investigation, a plan of the tissues in the stem was drawn.



Describe how a thin section of a stem could be prepared and viewed using a light microscope, in order to draw this plan.

(3)

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- (b) (i) The photograph shows an animal cell. This photograph was taken using an electron microscope.



(Source: © DR GOPAL MURTI/SCIENCE PHOTO LIBRARY)

Draw this cell as the student would see it under high power of a light microscope.

Label your diagram to show the nucleus, the nucleolus and the nuclear envelope.

(3)

- (ii) State **one** difference and **one** similarity between the nuclear envelope and the cell membrane.

(2)

Difference

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Similarity

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(Total for Question 1 = 16 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

